Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 1 of 11 · find part b in this mark scheme extract

5(a)	A1 A2	Award **two** marks for a linked explanation such as: • Replacing individual variables with an array (using a single name) (1) means that comparisons could be done in a loop (1) • The animals could be processed in a loop (1) because they could be stored in an array (under a single name) (1)		**2**

Question	mp	Answer	Additional guidance	Mark

Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 2 of 11 · find part b in this mark scheme extract

5(b)		Award **one** mark for each of:	Ignore contents of fourth row if given.	
	B1	First row high index and low index (1) • 9, 0	Accept • 20, 10	
	B2	First row mid point (1) • 4	Accept • 15	
	B3	Second row (1) • 3, 0, 1	Accept • 14, 10, 12	
	B4	Third row (1) • 0, 0, 0	Accept • 10, 10, 10	**4**

Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 3 of 11 · find part b in this mark scheme extract

Example: Using index

highIndex	lowIndex	midPoint
9	0	4
3	0	1
0	0	0

Example: Using Values

highIndex	lowIndex	midPoint
20	10	15
14	10	12
10	10	10

Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 4 of 11 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 5 of 11 · find part b in this mark scheme extract

5(c)		Award **one** mark for each of:		
	C1	Opens the file for reading (1)		
	C2	Reads each line from the file (1)	Requires both read lines (prime, inside loop) for C# and Java. Requires a foreach for Python. Allow FT from C1	
	C3	Splits each line into individual strings (1)	Syntactically correct <anyString>.split(',') Does not need to be splitting correct line	
	C4	Individual values cast to int for processing (1)	Any method (index or foreach) allow float no FT allowed from C1 to C3 – must be full values between commas in file	
	C5	SubTotal += value **and** grandTotal += subTotal (1)	Any reasonable attempt at calculating both totals	
	C6	All given variable and constant names used throughout (1)	INPUT_FILE COMMA subTotal grandTotal	
	C7	Subtotals displayed / Grand total displayed (1)	Must be accurate as shown in QP Fig 8 Not hard-coded in any way	
	C8	The input file is closed, before exiting (1)	Syntactically correct Must be file name originally opened – FT from C1	**8**

Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 6 of 11 · find part b in this mark scheme extract

Q	mp	Code Examples

Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 7 of 11 · find part b in this mark scheme extract

C#

```
// Q05cFINISHED

using System.IO;

String inputFile = "Sales.txt";     // Output file name
String comma = ",";                 // Use as a constant

int subTotal = 0;               // Subtotal for each line
int grandTotal = 0;             // Running total
String line;                    // Line read from file

// Complete the code to open the file for reading
String fullPath = "C:\\Q05c";
inputFile = fullPath + "\\" + inputFile;
StreamReader theFile = new StreamReader (inputFile);

// Complete the code to read the first line of the input file
line = theFile.ReadLine ();

// Loop as long as the line is not null
while (line != null)
{
    // Complete the code to split the line into a set of five strings
    string[] stringSales = line.Split (comma);

    subTotal = 0;
    // Add code to sum up each value in the set of five strings
    foreach (string sale in stringSales)
    {
        subTotal = subTotal + Convert.ToInt32 (sale);
    }

    // Add code to display the subtotal for the line
    Console.WriteLine (Convert.ToString (subTotal));

    // Add code to calculate the running total
    grandTotal = grandTotal + subTotal;

    // Complete the code to read the next line of the file
    line = theFile.ReadLine ();
}

// Add code to display the total of all lines in the file
Console.WriteLine ("Grand total: " + Convert.ToString (grandTotal));

// Complete the code to close the opened file
theFile.Close ();

```

Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 8 of 11 · find part b in this mark scheme extract

Java

```java
// Q05cFINISHED

import java.io.FileReader;
import java.io.BufferedReader;

public class Q05cFINISHED
{
    public static void main(String[] args) throws Exception
    {
        String inputFile = "Sales.txt";     // Output file name
        String comma = ",";                  // Use as a constant

        int subTotal = 0;                // Subtotal for each line
        int grandTotal = 0;              // Running total
        String line;                     // Line read from file

        // Complete the code to open the file for reading
        String full_path = "C:\\Q05c";
        inputFile = full_path + "\\" + inputFile;
        BufferedReader theFile = new BufferedReader (new FileReader (inputFile));

        // Complete the code to read the first line of the input file
        line = theFile.readLine();

        // Loop as long as the line is not null
        while (line != null)
        {
            // Complete the code to split the line into a set of five strings
            String[] stringSales = line.split (comma);

            subTotal = 0;
            // Add code to sum up each value in the set of five strings
            for (int i = 0; i < stringSales.length; i++)
            {
                subTotal = subTotal + Integer.parseInt (stringSales[i]);
            }

            // Add code to display the subtotal for the line
            System.out.println (String.valueOf (subTotal));

            // Add code to calculate the running total
            grandTotal = grandTotal + subTotal;

            // Complete the code to read the next line of the file
            line = theFile.readLine();
        }

        // Add code to display the total of all lines in the file
        System.out.println ("Grand total: " + String.valueOf (grandTotal));

        // Complete the code to close the opened file
        theFile.close ();
    }
}
```

Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 9 of 11 · find part b in this mark scheme extract

Python

```python
# Q05cFINISHED

INPUT_FILE = "Sales.txt"    # Output file name
COMMA = ","                 # Use as a constant

subTotal = 0        # Subtotal for each line
grandTotal = 0      # Running total

# Complete the code to open the file for reading
theFile = open (INPUT_FILE, "r")

# Complete the code to read each line of the input file
for line in theFile:

    # Complete the code to split the line into a set of five strings
    stringSales = line.split (COMMA)

    subTotal = 0
    # Add code to sum up each value in the set of five strings
    for sale in stringSales:
        subTotal = subTotal + int (sale)

    # Add code to display the subtotal for the line
    print (str (subTotal))

    # Add code to calculate the running total
    grandTotal = grandTotal + subTotal

# Add code to display the total of all lines in the file
print ( "Grand total: " + str (grandTotal))

# Complete the code to close the opened file
theFile.close ()

```

Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 10 of 11 · find part b in this mark scheme extract

Type MS · Binary search step by step

Answer 1 of 4 · Question 5 part b · 4 marks

2024 JUN · Paper 02 · extract 11 of 11 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Binary search step by step

Answer 2 of 4 · Question 4 part b.ii · 4 marks

2021 JUN · Paper 02 · extract 1 of 7 · find part b.ii in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
4 (b)(i)	B1	Award up to 2 marks for a linked explanation: • binary search can be quicker / is more effective than a linear search / is more effective with larger lists (1) as it does not have to examine each item in the list (1) • binary search halves the list each time (1) so it can be faster to find an item (1) • binary search requires fewer comparisons than a linear search to establish an item is not in the list (1) because the linear list would need to compare each item before establishing this (1)		(2)

Type MS · Binary search step by step

Answer 2 of 4 · Question 4 part b.ii · 4 marks

2021 JUN · Paper 02 · extract 2 of 7 · find part b.ii in this mark scheme extract

would need to compare each item before establishing this (1) (2)

4 (b)(ii) Correct answer

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	1
5	mil19	
6	rib10	2
7	str15	3
8	tor16	

Award one mark for each correct value in order column (4)

B1	Start of search correct	Accept 5 and 7 for B1 and B2 (2 marks)
B2	Second search item correct	Follow through if start of search incorrect
B3	Third search item correct	Follow through if start of search incorrect
B4	All correct	

Type MS · Binary search step by step

Answer 2 of 4 · Question 4 part b.ii · 4 marks

2021 JUN · Paper 02 · extract 3 of 7 · find part b.ii in this mark scheme extract

		All correct		
4 (b)(iii)	B5	Award 1 mark for: $\lceil \log_2 n \rceil + 1$		(1)

Type MS · Binary search step by step

Answer 2 of 4 · Question 4 part b.ii · 4 marks

2021 JUN · Paper 02 · extract 4 of 7 · find part b.ii in this mark scheme extract

		or $\log_2 7 + 1$		(1)
4 (b)(iv)	B6	Award 1 mark for any of: • bubble sort • merge sort	Accept any known sorting algorithm	(1)

Type MS · Binary search step by step

Answer 2 of 4 · Question 4 part b.ii · 4 marks

2021 JUN · Paper 02 · extract 5 of 7 · find part b.ii in this mark scheme extract

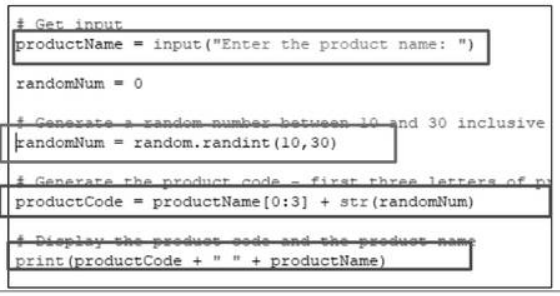
Python Code Mark Scheme

Question	mp	Answer	Additional Guidance	Mark

Type MS · Binary search step by step

Answer 2 of 4 · Question 4 part b.ii · 4 marks

2021 JUN · Paper 02 · extract 6 of 7 · find part b.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
4 (a)	A1	At least 1 variable has a meaningful name	Ignore spelling mistakes in input message	(7)
	A2	Product name requested using a suitable input message		
	A3	Random number generated that would be at least 10 **or** no higher than 30		
	A4	Random number generated that would be in the correct range 10 to 30		
	A5	First 3 letters of product name generated		
	A6	First 3 letters of product name and random number concatenated to generate productCode		
	A7	productCode and productName output in the same print statement		
Code examples				
Python				

```
# Get input
productName = input("Enter the product name: ")

randomNum = 0

# Generate a random number between 10 and 30 inclusive
randomNum = random.randint(10,30)

# Generate the product code - first three letters of p
productCode = productName[0:3] + str(randomNum)

# Display the product code and the product name
print(productCode + " " + productName)
```

Type MS · Binary search step by step

Answer 2 of 4 · Question 4 part b.ii · 4 marks

2021 JUN · Paper 02 · extract 7 of 7 · find part b.ii in this mark scheme extract

For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

Question	ran	Answer	Additional Guidance	Mark

Type MS · Binary search step by step

Answer 3 of 4 · Question 4 part b.iii · 1 mark

2021 JUN · Paper 02 · extract 1 of 7 · find part b.iii in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
4 (b)(i)	B1	Award up to 2 marks for a linked explanation: • binary search can be quicker / is more effective than a linear search / is more effective with larger lists (1) as it does not have to examine each item in the list (1) • binary search halves the list each time (1) so it can be faster to find an item (1) • binary search requires fewer comparisons than a linear search to establish an item is not in the list (1) because the linear list would need to compare each item before establishing this (1)		(2)

Type MS · Binary search step by step

Answer 3 of 4 · Question 4 part b.iii · 1 mark

2021 JUN · Paper 02 · extract 2 of 7 · find part b.iii in this mark scheme extract

would need to compare each item before establishing this (1) (2)

4 (b)(ii)

Correct answer

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	1
5	mil19	
6	rib10	2
7	str15	3
8	tor16	

Award one mark for each correct value in order column (4)

B1	Start of search correct	Accept 5 and 7 for B1 and B2 (2 marks)
B2	Second search item correct	Follow through if start of search incorrect
B3	Third search item correct	Follow through if start of search incorrect
B4	All correct	

Type MS · Binary search step by step

Answer 3 of 4 · Question 4 part b.iii · 1 mark

2021 JUN · Paper 02 · extract 3 of 7 · find part b.iii in this mark scheme extract

		All correct		
4 (b)(iii)	B5	Award 1 mark for: $2 \text{ or } \log_2 n + 1$		(1)

Type MS · Binary search step by step

Answer 3 of 4 · Question 4 part b.iii · 1 mark

2021 JUN · Paper 02 · extract 4 of 7 · find part b.iii in this mark scheme extract

		or $\log_2 7 + 1$		(1)
4 (b)(iv)	B6	Award 1 mark for any of: • bubble sort • merge sort	Accept any known sorting algorithm	(1)

Type MS · Binary search step by step

Answer 3 of 4 · Question 4 part b.iii · 1 mark

2021 JUN · Paper 02 · extract 5 of 7 · find part b.iii in this mark scheme extract

Python Code Mark Scheme

Question	mp	Answer	Additional Guidance	Mark

Type MS · Binary search step by step

Answer 3 of 4 · Question 4 part b.iii · 1 mark

2021 JUN · Paper 02 · extract 6 of 7 · find part b.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
4 (a)	A1	At least 1 variable has a meaningful name	Ignore spelling mistakes in input message	(7)
	A2	Product name requested using a suitable input message		
	A3	Random number generated that would be at least 10 **or** no higher than 30		
	A4	Random number generated that would be in the correct range 10 to 30		
	A5	First 3 letters of product name generated		
	A6	First 3 letters of product name and random number concatenated to generate productCode		
	A7	productCode and productName output in the same print statement		
Code examples				
Python				

```
# Get input
productName = input("Enter the product name: ")

randomNum = 0

# Generate a random number between 10 and 30 inclusive
randomNum = random.randint(10,30)

# Generate the product code - first three letters of p
productCode = productName[0:3] + str(randomNum)

# Display the product code and the product name
print(productCode + " " + productName)
```

Type MS · Binary search step by step

Answer 3 of 4 · Question 4 part b.iii · 1 mark

2021 JUN · Paper 02 · extract 7 of 7 · find part b.iii in this mark scheme extract

For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

Question	ran	Answer	Additional Guidance	Mark

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

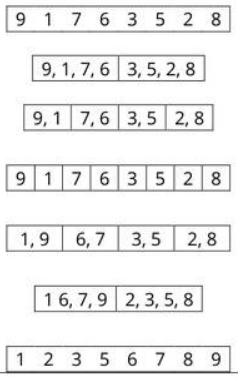
2021 NOV · Paper 02 · extract 1 of 18 · find part b in this mark scheme extract

2 (b)	B1 B2 B3 B4	**Award up to 4 marks for a linked explanation such as:** • The list is not sorted (1) Juan should come after Elija if the list was sorted / Elija is compared with Juan (1), so the bottom half of the list would be discarded (1) after the first pass through the loop (1) • Binary search works on a sorted list (1) because it uses divide and conquer (1) half the list discarded each pass through (1) based on comparison of search item with middle item (1)		(4)
Question	**mp**	**Answer**	**Additional Guidance**	**Mark**

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 2 of 18 · find part b in this mark scheme extract

2 (c)	C1 C2 C3 C4 C5	**Award 1 mark for each of:** Split into two sets of four (1) Split each set into two pairs (1) Split each pair into single elements (1) Merge elements into sorted pairs (1) Merge pairs into sorted sets (1) 	Allow step at C3 to be implied	(5)

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 3 of 18 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 4 of 18 · find part b in this mark scheme extract

2 (a)	**Award one mark for each of:**		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the input message.	
	A1	Input as an integer (1)		
	A2	Loop with correct conditions start <= end and not found (1)		
	A3	Use of correct symbol for integer division (in conversion of DIV) or type coercion (1)		
	A4	Check to see if number in middle matches item (1)		
	A5	Set found to true (1)		
	A6	Check to see if item is less than number in middle (1)		
	A7	Set end to middle – 1 (1)		
	A8	else – set start to middle + 1 (1)		
	A9	count incremented by 1 (1)		
	A10	Program execution is fully correct (1)		(10)
Code examples				

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 5 of 18 · find part b in this mark scheme extract

C#

```
Console.WriteLine("What is the number to find? ");
int item = Convert.ToInt32(Console.ReadLine());

while (start <= end && !found)
{
    middle = (start + end) / 2;
    if (numberList[middle] == item)
    {
        found = true;
    }
    else
    {
        if (item < numberList[middle])
        {
            end = middle - 1;
        }
        else
        {
            start = middle + 1;
        }
    }
    count++;
}
```

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 6 of 18 · find part b in this mark scheme extract

Java

```
Scanner input = new Scanner(System.in);
System.out.print("What is the number to find? ");
int item = Integer.parseInt(input.nextLine());

while (start <= end && !found)
{
    middle = (start + end) / 2;
    if (numberList[middle] == item)
    {
        found = true;
    }
    else
    {
      if (item < numberList[middle])
      {
          end = middle - 1;
      }
      else
      {
          start = middle + 1;
      }
    }
    count ++;
}
```

Python

```
item = int(input("What is the number to find? "))
while start <= end and not found:
    middle = (start + end) // 2
    if numberList[middle] == item:
        found = True
    else:
        if item < numberList[middle]:
            end = middle - 1
        else:
            start = middle + 1
    count = count + 1
```

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 7 of 18 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 8 of 18 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
4 (a)	**Award one mark for each of:**			
		Validation		
	A1	Suitable input prompt and storing it in binaryPattern (1)		
	A2	Length of binary pattern validated as 8 (1)		
	A3	Loop used to check the length of binary pattern (1)		
		Binary to denary		
	A4	Loop through each digit in the binary pattern (1)		
	A5	Correct conversion of at least one binary digit to relevant placeholder (1)		
	A6	Attempt at running total using denaryNumber (1)		
	A7	Correct running total using denaryNumber (1)		
	A8	Print statement includes the binary pattern and the denary number (1)		
	A9	Print statement outside of conversion loop (1)		
	A10	Comment explains how conversion works (1)		
	A11	Program executes correctly for any binary pattern that is exactly 8 characters long only (1)		(11)
Code examples				

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 9 of 18 · find part b in this mark scheme extract

C#

```
while (binaryPattern.Length != 8)
{
    Console.WriteLine("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = Console.ReadLine();
}

foreach (char digit in binaryPattern)
{
    if (digit == '1')
    {
        denaryNumber += denaryPlaceholders[count] * 1;
    }
    count++;
}
Console.WriteLine(binaryPattern + " converted to denary is " + denaryNumber);
Console.ReadLine();
```

Java

```
while (binaryPattern.length() != 8)
{
    Scanner input = new Scanner(System.in);
    System.out.print("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = input.next();
}

for (char digit : binaryPattern.toCharArray())
{
    if (digit == '1')
    {
       denaryNumber += denaryPlaceholders[count] * 1;
    }
    count ++;
}
System.out.print(binaryPattern + " converted to denary is " + denaryNumber);
```

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 10 of 18 · find part b in this mark scheme extract

Python

```
while len(binaryPattern) != 8:
    binaryPattern = input("Enter an 8 digit binary number to convert to denary: ")

for digit in binaryPattern:
    if digit == "1":
        denaryNumber = denaryNumber + denaryPlaceholders[count] * 1
    count = count + 1

print(binaryPattern,"converted to denary is", denaryNumber)
```

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 11 of 18 · find part b in this mark scheme extract

For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit

Question	mp	Answer	Additional Guidance	Mark
5	A1	At least three variables initialised appropriately		(1)
	A2	Word input		(1)
	A3	Program repeats until 1 is input		(1)
	A4	Loop through each word in the array		(1)
		Words that begin with the same letter		
	A5	Check to see if the first letter of the input word matches the first letter of at least one word in the array		(1)
	A6	The number of words that begin with the same letter displayed		(1)
		Words that contain the input word		
	A7	Check to see if the word contains the input word		(1)
	A8	Number of letters in the word calculated		(1)
	A9	At least one word from the array that contains the input word displayed		(1)
	A10	The number of words that contain the input word displayed		(1)
	A11	Identification of the longest word or the shortest word		(1)

Band 1 (1-3 marks)	Band 2 (4-6 marks)	Band 3 (7-9 marks)	Mark
Little attempt to decompose into component parts	Some attempt to decompose into component parts	The problem has been decomposed into component parts	
Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	(9)

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 12 of 18 · find part b in this mark scheme extract

Code examples

C#

```
// Add your code here
string inputWord = "";

while (inputWord != "1")
{
    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    Console.WriteLine("Enter a word or 1 to exit ");
    inputWord = Console.ReadLine();

    if (inputWord != "1")
    {
        int count = 0;

        foreach (String word in wordArray)
        {
            if(word[0] == inputWord[0])
            {
                Console.WriteLine(word);
                count ++;
            }
        } // End of loop

        Console.WriteLine(count + " word(s) beginning with "+ inputWord[0]);
        Console.WriteLine("----------------------------------------------------------------------");

        count = 0;

        foreach (String word in wordArray)
        {
            if(word.Contains(inputWord))
            {
                count ++;
                Console.WriteLine(word);

                int length = word.Length;

                if (longest < length)
```

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 13 of 18 · find part b in this mark scheme extract

```
                    {
                        longest = length;
                        longestWord = word;
                    }

                    if (shortest > length)
                    {
                        shortest = length;
                        shortestWord = word;
                    }
                }// end of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word

            if (count > 0)
            {
                Console.WriteLine(count + " word(s) with "+ inputWord + " in them");
                Console.WriteLine("The longest word has " + longest + " characters");
                Console.WriteLine("The shortest word has " + shortest + " characters");
                Console.WriteLine("The longest word is " + longestWord + " characters");
                Console.WriteLine("The shortest word is " + shortestWord + " characters");
                Console.WriteLine("--------------------------------------------------------------------------------");
            }
            else
            {
                Console.WriteLine("There were 0 words that had all the letters from " + inputWord + " in them");
                Console.WriteLine("--------------------------------------------------------------------------------");
            } // End of printing words
        } // End of check
    } // End of while loop
```

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 14 of 18 · find part b in this mark scheme extract

Java

```
// Add your code here
Scanner input = new Scanner(System.in);
String inputWord = "";

// While there is a word run the program
while (!inputWord.equals("1")) {

    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    System.out.println("Enter a word or 1 to exit ");
    System.out.println("--------------------------------------------------------------------------------");
    inputWord = input.nextLine();
    if (!inputWord.equals("1")){
        // Find words that begin with the same letter as the input word
        int count = 0;

        // Get each word in the array
        for (String word: wordArray){

            // If the first character of the word is the same as the first character of
            // the input word then print the word
            if (word.charAt(0) == inputWord.charAt(0)){
                System.out.println(word);
                count ++;
            }
        } // End of for loop

        // Print the total number of words that begin with the same letter
        System.out.println(count + " word(s) beginning with "+ inputWord.charAt(0));
        System.out.println("--------------------------------------------------------------------------------");

        // Find words that contain the input word
        count = 0;

        // Get each word in the array
        for (String word: wordArray){
            // Check to see if the word contains the inputWord
            if (word.contains(inputWord)){
                count ++;
                System.out.println(word);

                // Set the length of the word
                int length = word.length();
```

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 15 of 18 · find part b in this mark scheme extract



```
                    // If the length is longer than the current longest replace it and the longestWord
                    if (longest < length) {
                        longest = length;
                        longestWord = word;
                    }

                    // If the length is shorter than the shortest replace it and the shortestWord
                    if (shortest > length){
                        shortest = length;
                        shortestWord = word;
                    }
                } // End of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word
            if( count > 0){
                System.out.println(count + " word(s) with "+ inputWord + " in them");
                System.out.println("The longest word has " + longest + " characters");
                System.out.println("The shortest word has " + shortest + " characters");
                System.out.println("The longest word is " + longestWord + " characters");
                System.out.println("The shortest word is " + shortestWord + " characters");
                System.out.println("--------------------------------------------------------------------------------");
            } else {
                System.out.println("There were 0 words that had all the letters from " + inputWord + " in them");
                System.out.println("--------------------------------------------------------------------------------");
            } // End of printing words
        }
    } // End of while loop
```

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 16 of 18 · find part b in this mark scheme extract

Python

```python
# While there is a word run the program
while inputWord !="1":

    shortest = 5000
    longest = 0
    shortestWord = ""
    longestWord = ""

    inputWord = input("Enter a word or 1 to exit: ")
    print("------------------------------------------------------------------------")

    if inputWord !="1":
        # Find words that begin with the same letter as the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # If the first character of the word is the same as the first character of
            # the input word then print the word
            if word[0] == inputWord[0]:
                print(word)
                count = count + 1

        # Print the total number of words that begin with the same letter
        print("\n",count,"word(s) beginning with",inputWord[0])
        print("------------------------------------------------------------------------")

        # Find words that contain the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # Check to see if the word contains the inputWord
            if inputWord in word:
                count = count + 1
                print(word)
                # Set the length of the word
                length = len(word)

                # If the length is longer than the current longest replace it and the longestWord
                if longest < length:
                    longest = length
                    longestWord = word

                # If the length is shorter than the shortest replace it and the shortestWord
                if shortest > length:
                    shortest = length
                    shortestWord = word
```

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

2021 NOV · Paper 02 · extract 17 of 18 · find part b in this mark scheme extract

```
59
60          # Print the number of words, the number of characters in the longest and
61          # shortest word, the longest and the shortest word
62          if count > 0:
63              print("\n",count,"word(s) with",inputWord,"in them")
64              print("The longest word has",longest,"letters")
65              print("The shortest word has",shortest,"letters")
66              print("The longest word is",longestWord)
67              print("The shortest word is",shortestWord)
68              print("-------------------------------------------------------------------------")
69          else:
70              print("There were 0 words with all the letters from",inputWord,"in them")
71          # ------------------------------------------------------------------------------------
72
73  # End the program when there are no more words
74  print("End of program")
```

Type MS · Binary search step by step

Answer 4 of 4 · Question 2 part b · 4 marks

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